Tools needed: ESP32 token virtual HID bridge on Windows

Companion to windows-cmd (build.cmd, install.cmd, start-bridge.cmd, setup-token.cmd). Nothing here has been run on a real token yet.

1. To build

Tool	Used for	Where to get it / notes
winlibs MinGW-w64, x86_64 build (gcc)	build.cmd tools: pico-vhid.exe, pico-install.exe	winlibs.com. Unzip, add its `bin` folder to PATH. Must be the 64-bit (x86_64) build.
Visual Studio 2022 (Community is fine) with workload "Desktop development with C++"	build.cmd driver: cl.exe and link.exe (x64)	visualstudio.microsoft.com/downloads. build.cmd finds it with vswhere, or run from the "x64 Native Tools Command Prompt".
Windows 10/11 SDK	headers, signtool.exe, and the base for the WDK	Tick it in the Visual Studio workload, or install the standalone SDK. Note its version, e.g. 10.0.26100.0.
Windows Driver Kit (WDK), **same version number as the SDK**	KMDF and VHF headers/libs (vhf.h, vhfkm.lib), stampinf.exe, Inf2Cat.exe	learn.microsoft.com/windows-hardware/drivers/download-the-wdk. Install the SDK first, then the WDK. The Visual Studio extension that the WDK offers is not needed. Check: `Windows Kits\10\Include\<ver>\km\vhf.h` must exist.
Windows PowerShell 5.1 (built into Windows)	only as a fallback: creating the test certificate if the WDK has no makecert.exe, or exporting a .cer from a .cat	Nothing to install.

2. To install and run

Tool / setting	Used for	Where to get it / notes
Windows 10 1709 or newer, 64-bit (Windows 11 works)	the driver targets 1709+ (build 16299)	Install, uninstall and start-bridge need an **administrator** Command Prompt.
Test-signing mode (bcdedit /set testsigning on) and **Secure Boot off**	loading the test-signed pico_vhf.sys	install.cmd offers to set it; needs a reboot. Turn it off with uninstall.cmd when done.
Python 3 with the `py` launcher	ctaphid_serial_bridge.py and fpctl.py (no extra packages)	python.org/downloads/windows. Tick "Add python.exe to PATH".
libfido2 tools for Windows (fido2-token.exe)	setup-token.cmd options 5 and 6, check-token.cmd	developers.yubico.com/libfido2/Releases/ . Put the folder with fido2-token.exe and its DLLs on PATH. Not needed for fingerprints only.
ESP-IDF v5.5 (ESP-IDF Tools Installer)	building and flashing the ESP32-C6 firmware	docs.espressif.com, Get Started for ESP32-C6 on Windows. Flash through the UART port.
USB data cable on the board's **native USB** port	the token link (usbser.sys, nothing to install)	A charge-only cable shows no COM port.
Built into Windows: cmd, certutil, bcdedit, reg, fltmc, taskkill, findstr, find, ping	the .cmd scripts	Nothing to install.

3. Order

1) SDK, then WDK (same version). 2) Visual Studio C++ workload. 3) winlibs on PATH. 4) `build.cmd tools`, then `build.cmd driver`. 5) As administrator: `install.cmd`, reboot if asked, run it again. 6) `setup-token.cmd` for fingerprints, then `start-bridge.cmd`.

Quick checks

`gcc -dumpmachine` prints x86_64-w64-mingw32 | `dir "%ProgramFiles(x86)%\Windows Kits\10\Include\*\km\vhf.h"` lists a file | `py -3 --version` works | `fido2-token -V` works (optional).